Electric fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use field arrows to predict the force on a test charge.

SUCCESS CRITERIA - I can...

1. Draw electric field arrows for isolated positive and negative charges.
2. Use the positive-test-charge convention.
3. Distinguish the field from a particular test charge's force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

In separate cases, an ideal blackbody temperature becomes 3 times as large and a sound frequency becomes 5 times as large at fixed sound speed. Compare the two wavelength changes.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $E = F/q$; $F = qE$.
Use consistent signed components. The field points along the force on a positive test charge; negative charge reverses force direction.

Worked reasoning: At a point where E=4 N/C right, a +2 C test charge feels 8 N right. A-2 C charge feels 8 N left. Field direction is defined using positive test charge.

Your turn: Draw the field arrow once and the two alternative force arrows.

Field / first contribution

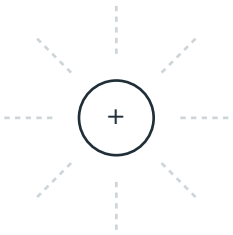
Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch field arrows around a positive source charge. Place a negative test charge to its right and draw that charge's force.



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Apply the physics

A tiny charged bead has charge -2.0×10^{-6} C in a uniform electric field of 300 N/C east. Calculate its electric force. Use separate arrows to explain why force and field point in different directions.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

05 Explain your reasoning

If you remove the test charge, does the source charge’s field disappear? Explain the role of a test charge.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A closed-open pipe has length 0.4 m. Use λ=4 L for the fundamental and v=340 m/s. Find its fundamental frequency.
Sketch the displacement pattern and label the pipe length.

closed-open tube



Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3

07 Exit check

At twice the distance from a point charge, how does field magnitude change? State the model assumption.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A -3 microcoulomb charge feels a 0.012 N force upward. Find the electric-field vector.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3